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Enable GPU Acceleration in Kubernetes Using NVIDIA GPU Operator

4 min read · Mar 28, 2025

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Introduction

As AI and machine learning applications grow in scale and complexity, the demand for powerful GPU resources is higher than ever. Kubernetes provides a scalable way to manage GPU workloads across your cluster efficiently

> ***Note:*** *You don't need to install NVIDIA drivers or CUDA manually — the operator handles it for you by deploying the necessary components across your GPU nodes.*

Step 1: Install Helm on the Master Node

Helm is a tool that helps you install and manage pods and other resources in Kubernetes. It uses something called “charts,” which are like pre-made setup files that make deployment easier and more consistent.

We’ll use Helm to install the GPU Operator and related components.

To install Helm, run the below commands:

```
$ curl -fsSL -o get_helm.sh https://raw.githubusercontent.com/helm/helm/master/
$ chmod 700 get_helm.sh
$ ./get_helm.sh
```

Step 2: Install the NVIDIA GPU Operator

The NVIDIA GPU Operator simplifies GPU management in Kubernetes. It includes essential components like the Node Feature Discovery (NFD) and the NVIDIA Driver Manager. GPU Operator enables faster node provisioning by automatically installing all necessary software components for running GPU-accelerated applications.

i. Create the GPU Operator Namespace

Begin by creating a new namespace for the GPU operator:

```
$ kubectl create ns gpu-operator
```

ii. Configure Pod Security Admission (Optional)

If your cluster uses Pod Security Admission (PSA), label the namespace to set the security policy to privileged:

```
$ kubectl label --overwrite ns gpu-operator pod-security.kubernetes.io/enforce=
```

iii. Check for Node Feature Discovery (NFD)

By default, Node Feature Discovery (NFD) is automatically deployed in the master and worker by the Nvidia Operator. If NFD is already running in the cluster, then

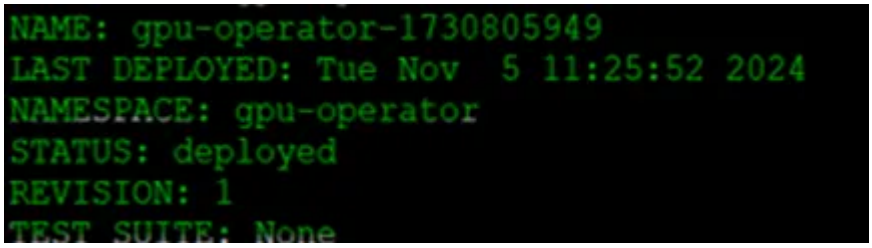
you must disable deploying NFD when you install the Operator

If NFD is already running on your cluster, you can skip deploying it again. To check if NFD is already deployed, use:

```
$ kubectl get nodes -o json | jq '.items[].metadata.labels | keys | any(startsw
```

iv. Add the NVIDIA Helm repository and install the GPU operator:

```
$ helm repo add nvidia https://helm.ngc.nvidia.com/nvidia && helm repo update
$ helm install --wait --generate-name \
  -n gpu-operator --create-namespace \
  nvidia/gpu-operator \
  --version=v25.3.0
```



Helm deployment status of the GPU Operator, confirming that it is deployed in the `gpu-operator` namespace

Step 3: Verify GPU Operator Deployment

To verify that the GPU operator was set up successfully, check the pods in the `**gpu-operator**` namespace:

```
$ kubectl get pods -n gpu-operator
```

```
root@k8master:/home/ubuntu# kubectl get pods -n gpu-operator
NAME                                                          READY   STATUS      RESTARTS   AGE
gpu-feature-discovery-6wtq5                                   1/1     Running     0          9m46s
gpu-operator-1731315579-node-feature-discovery-gc-bdcb6c5cgbc6w   1/1     Running     0          10m
gpu-operator-1731315579-node-feature-discovery-master-84967mttw   1/1     Running     0          10m
gpu-operator-1731315579-node-feature-discovery-worker-4hbk2   1/1     Running     0          10m
gpu-operator-1731315579-node-feature-discovery-worker-tq5nz   1/1     Running     0          10m
gpu-operator-79f94f69df-zrvrd                                 1/1     Running     0          10m
nvidia-container-toolkit-daemonset-7rfsf                      1/1     Running     0          9m47s
nvidia-cuda-validator-gj8g4                                   0/1     Completed   0          9m7s
nvidia-dcgm-exporter-zhfws                                    1/1     Running     0          9m47s
nvidia-device-plugin-daemonset-2p72t                          1/1     Running     0          9m47s
nvidia-operator-validator-glh6j                               1/1     Running     0          9m47s
```

Pods running inside ‘gpu-operator’ namespace

If all pods are running without errors, your GPU setup is complete!

Step 4: Test GPU Utilization with a Sample CUDA Application

To confirm that GPUs are correctly configured and can be utilized by containers, let’s deploy a sample CUDA application that performs a simple vector addition.

i. Create a `cuda-vectoradd.yaml` file

Save the following sample YAML configuration for a CUDA application:

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: cuda-vectoradd
spec:
  restartPolicy: OnFailure
  containers:
  - name: cuda-vectoradd
    image: "nvcr.io/nvidia/k8s/cuda-sample:vectoradd-cuda11.7.1-ubuntu20.04"
    resources:
      limits:
        nvidia.com/gpu: 1
```

ii. Deploy the Pod

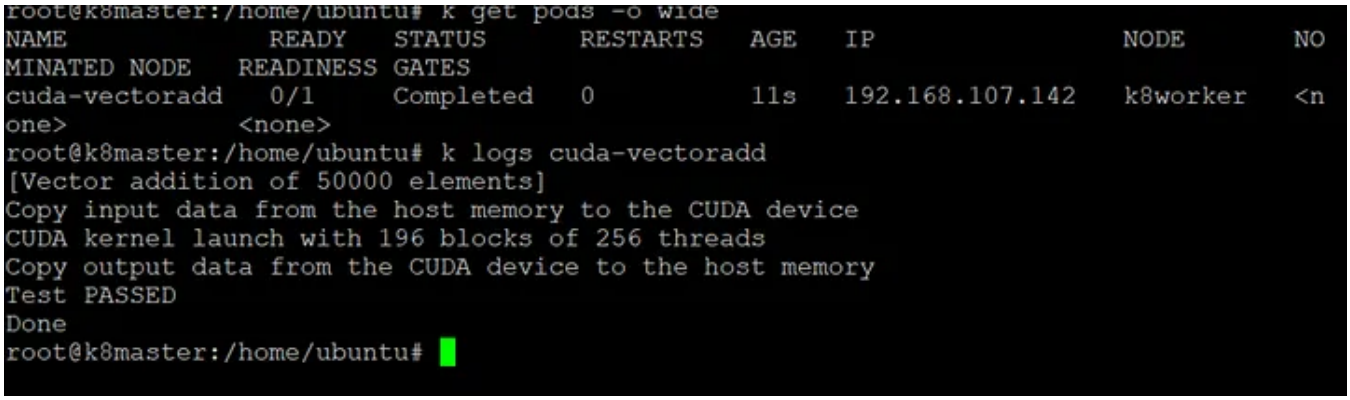
Run the following command to deploy the pod:

```
$ kubectl apply -f cuda-vectoradd.yaml
```

iii. View Logs to Confirm GPU Utilization

After the pod runs, check the logs to ensure it completed successfully:

```
$ kubectl logs cuda-vectoradd
```



Output from `kubectl logs` command for the `cuda-vectoradd` pod, showing the results of a sample GPU workload to verify CUDA functionality.

The output should indicate successful vector addition, verifying that the GPU works correctly.

Conclusion

With GPU support enabled through the NVIDIA GPU Operator, your Kubernetes cluster is now fully equipped to run GPU-accelerated AI and machine learning workloads. This setup not only simplifies GPU management but also enables scalable, high-performance computing for modern data-driven applications.

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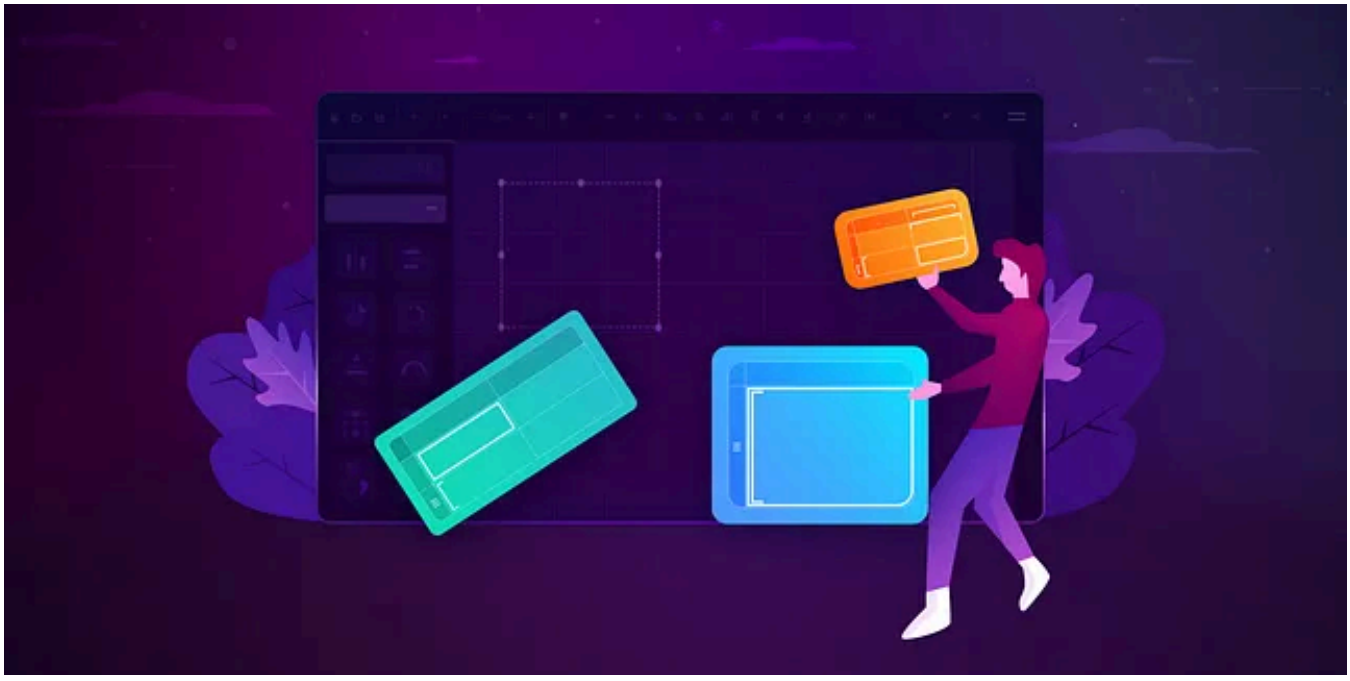

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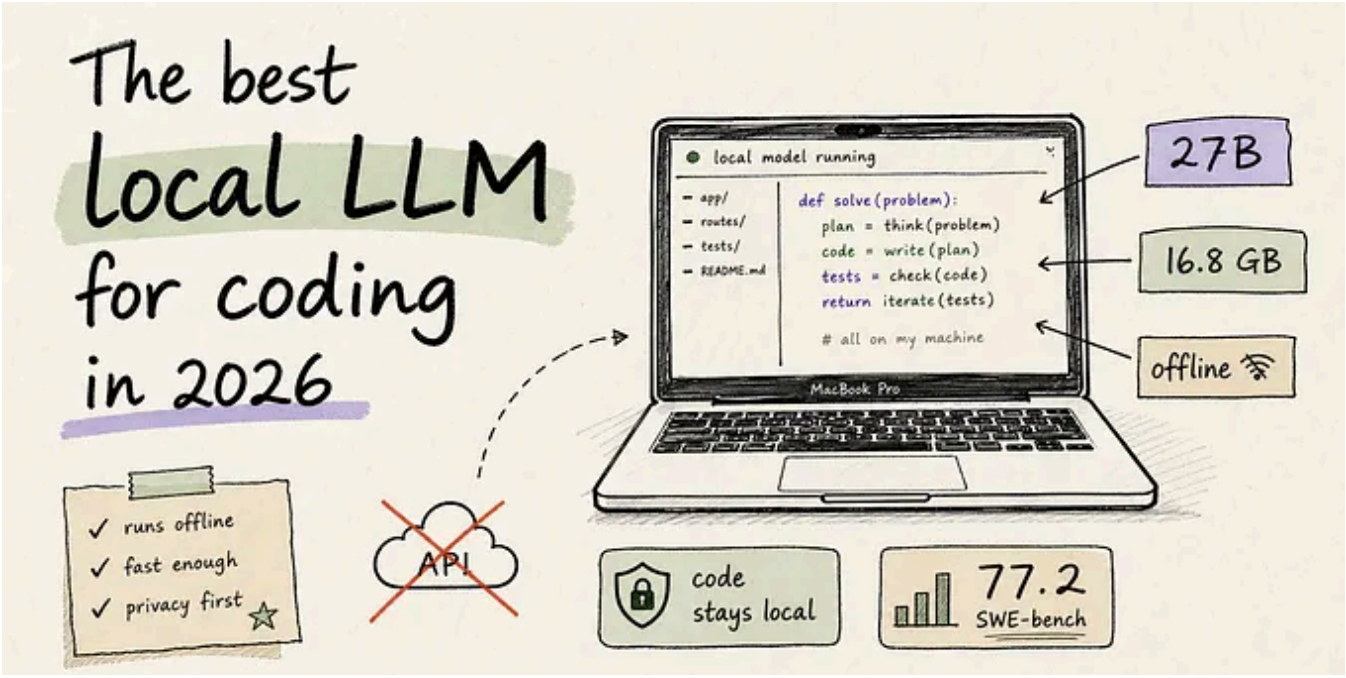

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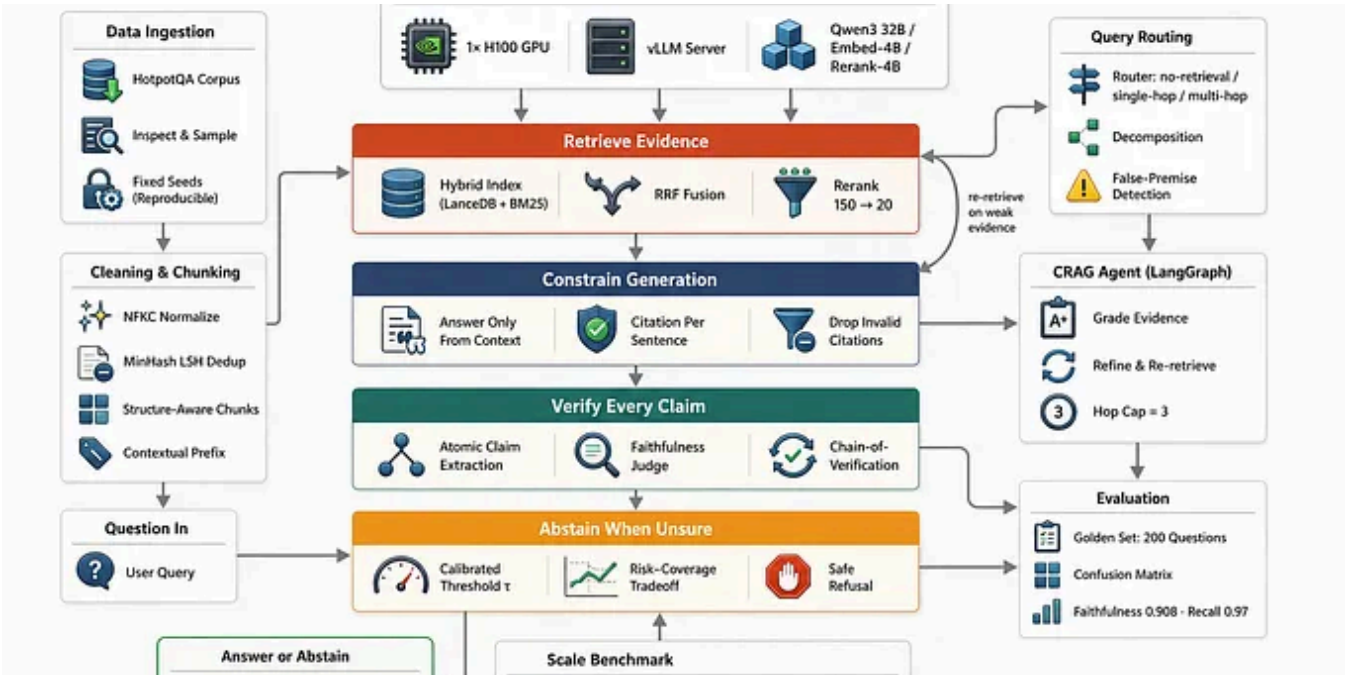

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Component	Precision	Approximate Size
MoE routed experts (384 per layer x 60 layers)	INT4	~400GB
Attention / MLA (61 layers)	BF16	~120GB
Shared experts (60 layers)	BF16	~10GB
Dense MLP (layer 0)	BF16	~5GB
Embeddings + lm_head	BF16	~14GB
Total weight memory		**~549GB**

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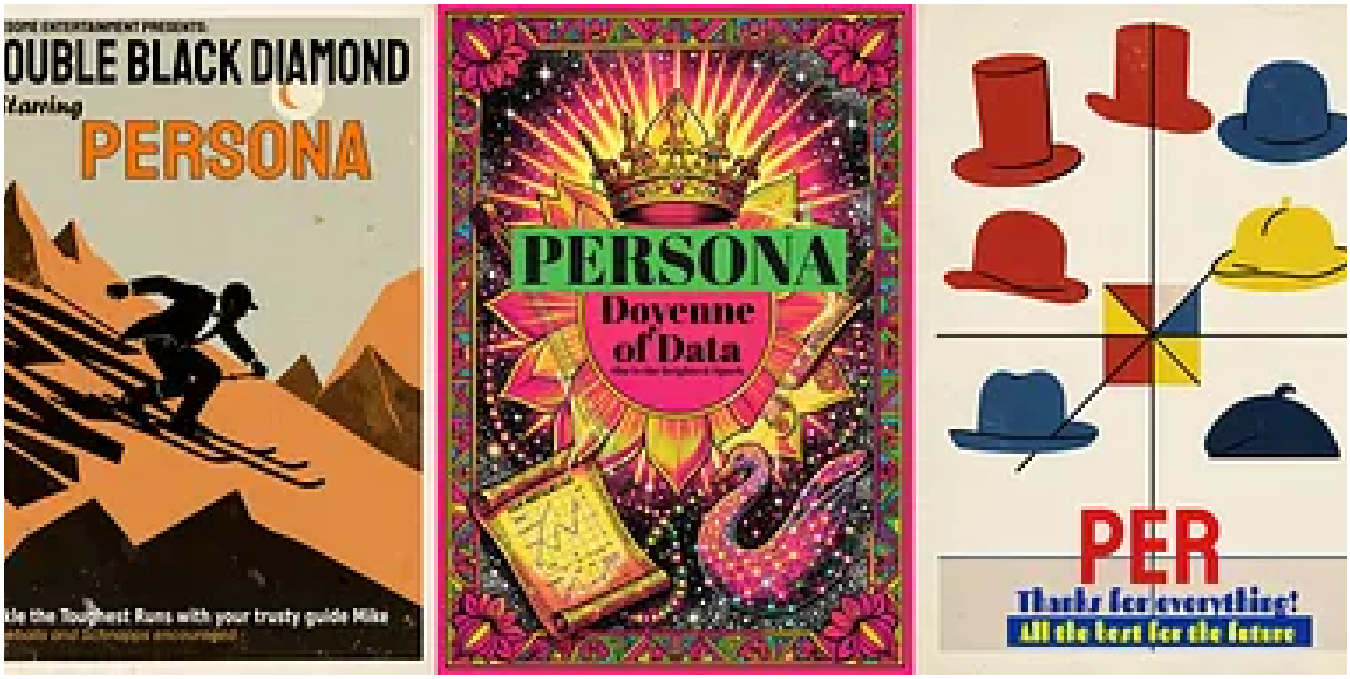

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